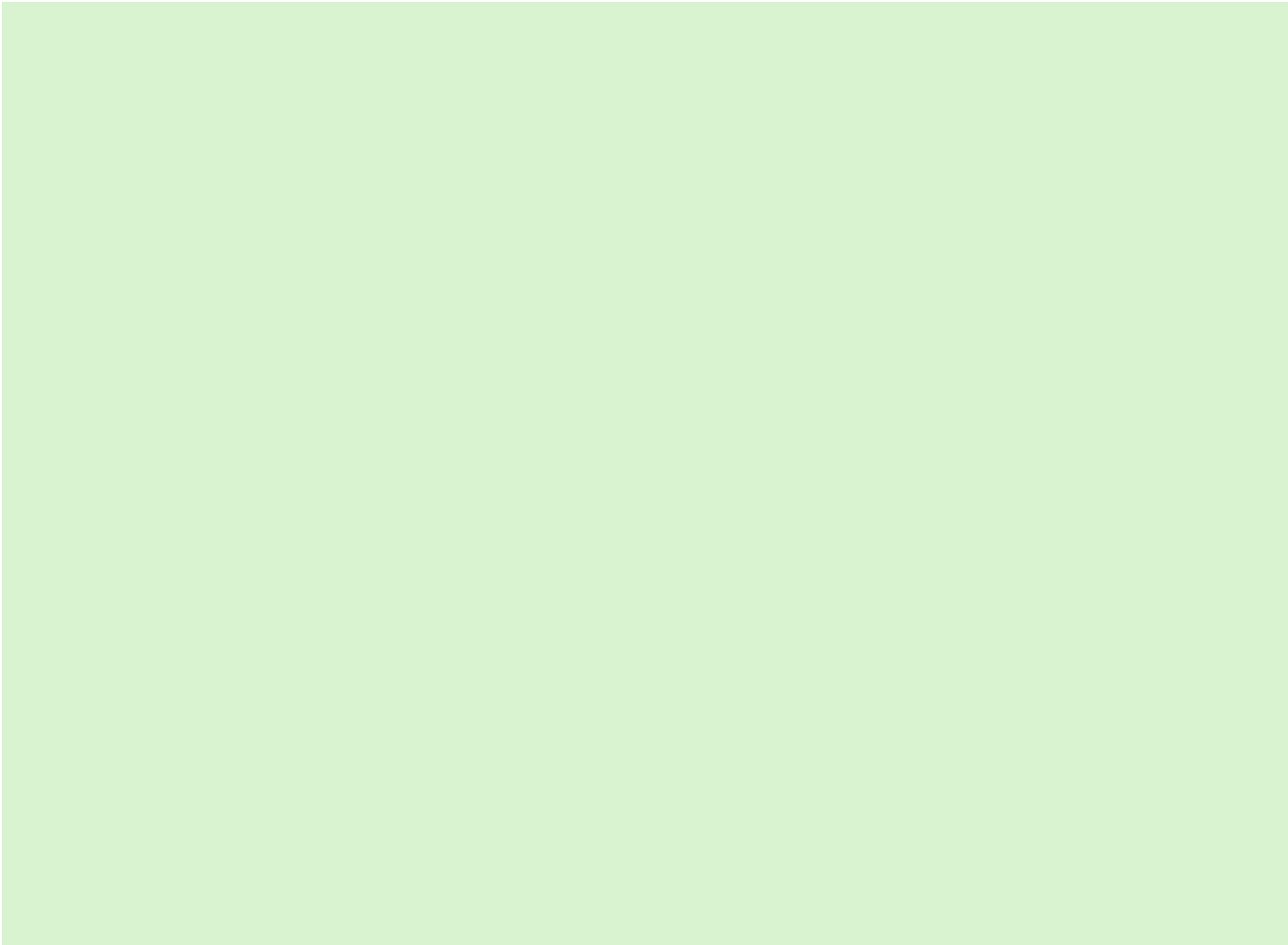




FINAL PROJECT REPORT



NEXTFORM : WASTE TRANSFORMATION & OPPORTUNITY PLATFORM

FINAL PROJECT REPORT

Project Description

NextForm is an AI-assisted Waste Transformation & Opportunity Platform developed to encourage responsible consumption by helping users discover innovative ways to transform discarded materials into useful products. The platform promotes reuse, supports circular economy practices, highlights sustainability impacts, and explores business opportunities from waste materials.

Submitted By

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Team Name: Green Guardians

Internship: 1M1B AI for Sustainability Virtual Internship

In Collaboration With: IBM SkillsBuild & AICTE

Acknowledgement

We express our sincere gratitude to **1M1B Foundation**, **IBM SkillsBuild**, and **AICTE (All India Council for Technical Education)** for organizing the **1M1B AI for Sustainability Virtual Internship** and providing us with the opportunity to explore the application of Artificial Intelligence in addressing real-world sustainability challenges.

This internship has been an enriching learning experience that enhanced our understanding of AI concepts, prompt engineering, responsible AI, and the United Nations Sustainable Development Goals (SDGs). The knowledge and practical exposure gained throughout the program played a significant role in the successful development of our project, **NextForm – Waste Transformation & Opportunity Platform**.

We would also like to express our heartfelt appreciation to our mentors, resource persons, and the IBM SkillsBuild learning platform for their valuable guidance, technical resources, and continuous support throughout the internship.

Our sincere thanks go to our college management and faculty members for their encouragement and support during the completion of this project.

Finally, we extend our gratitude to each other as team members for our dedication, collaboration, and commitment throughout the planning, development, testing, and successful deployment of the project.

We are grateful to everyone who directly or indirectly contributed to the successful completion of this project.

Sreelekshmi A A

Sooryajith S

Team Green Guardians

Abstract

The rapid increase in waste generation has become one of the most significant environmental challenges worldwide. A large quantity of reusable materials such as plastic, glass, paper, fabric, metal, and electronic waste is discarded every day due to limited awareness of sustainable reuse opportunities. This contributes to environmental pollution, resource depletion, and increased pressure on landfills.

NextForm – Waste Transformation & Opportunity Platform is an AI-assisted sustainability platform developed to encourage responsible consumption and promote circular economy practices. The platform enables users to enter the name of a discarded item and receive practical upcycling suggestions, material identification, estimated business opportunities, sustainability benefits, and alignment with the United Nations Sustainable Development Goals (SDGs).

The project has been developed using **Python, Streamlit, IBM SkillsBuild AI concepts, IBM Granite Models** (*where applicable to the implementation*), **HTML, CSS, and GitHub**. The application has been deployed on **Streamlit Community Cloud**, making it accessible through a public website.

The primary objective of NextForm is to reduce waste by helping individuals recognize the hidden value of discarded materials and encouraging innovative reuse instead of disposal. The platform supports **SDG 12 – Responsible Consumption and Production** by promoting sustainable decision-making, environmental awareness, and responsible resource utilization.

Through this project, the team demonstrates how AI concepts can be applied to address real-world sustainability challenges while encouraging environmental responsibility, green innovation, and community participation. NextForm serves as a simple yet effective solution that bridges technology and sustainability, inspiring users to transform waste into valuable opportunities.

Introduction

Background of the Study

The rapid growth of urbanization, industrialization, and consumer lifestyles has significantly increased the amount of waste generated worldwide. A considerable portion of this waste consists of materials such as plastic, glass, paper, fabric, metal, and electronic components that still possess reuse potential. However, due to limited awareness and inadequate guidance, these materials are often discarded, contributing to environmental pollution and resource depletion.

The concept of a **circular economy** encourages extending the lifecycle of products through reuse, repair, recycling, and upcycling instead of disposal. Artificial Intelligence (AI) has emerged as a powerful technology capable of assisting individuals in making sustainable decisions by providing intelligent recommendations and insights.

NextForm – Waste Transformation & Opportunity Platform was developed to promote sustainable waste management by helping users discover innovative ways to transform discarded materials into useful products. The platform encourages responsible consumption while creating awareness about environmental sustainability.

Need for Sustainable Waste Management

Improper waste disposal has become a major environmental concern due to the increasing volume of municipal and household waste. Valuable resources are frequently lost because reusable materials are discarded without considering their transformation potential.

There is a growing need for accessible digital solutions that encourage individuals to adopt sustainable practices. By providing practical reuse ideas and highlighting the environmental and economic value of waste transformation, NextForm aims to reduce landfill waste, conserve natural resources, and promote circular economy principles.

Artificial Intelligence for Sustainability

Artificial Intelligence is transforming the way environmental challenges are addressed by enabling intelligent decision-making, automation, and data-driven recommendations. AI can assist users in identifying materials, generating innovative reuse ideas, and providing sustainability insights that support environmentally responsible behavior.

The concepts learned through the IBM SkillsBuild AI for Sustainability Internship were applied during the development of NextForm to demonstrate how AI can be used as a tool for promoting sustainability and responsible consumption.

Motivation of the Project

The inspiration for NextForm originated from observing that many useful materials are discarded simply because people are unaware of their reuse potential. While recycling initiatives exist, there is limited awareness regarding creative upcycling opportunities that can extend the life of discarded objects.

Our motivation was to develop a simple, user-friendly platform that encourages individuals to view waste as a valuable resource rather than as something to be thrown away. The project also provided an opportunity to apply AI concepts learned during the internship to solve a real-world sustainability challenge.

Project Overview

NextForm is an AI-assisted Waste Transformation & Opportunity Platform that recommends practical reuse ideas for discarded materials. Users can enter the name of an item through the web interface, and the platform provides material identification, innovative transformation suggestions, estimated business opportunities, sustainability impacts, and alignment with **Sustainable Development Goal 12 – Responsible Consumption and Production**.

The application was developed using **Python, Streamlit, Visual Studio Code, GitHub, and Streamlit Community Cloud** for deployment. By combining technology with sustainability, NextForm aims to encourage responsible consumption, reduce waste generation, and promote environmental awareness through an intuitive and accessible platform.

Problem Statement

Existing Problem

The increasing generation of waste has become one of the most critical environmental issues worldwide. Every day, households, educational institutions, and industries generate large quantities of waste materials such as plastic bottles, glass containers, paper, cardboard, fabric, and metal. Although many of these materials are reusable, they are often discarded due to a lack of awareness about their potential value.

Most individuals are unfamiliar with creative upcycling methods or sustainable alternatives to disposal. As a result, reusable materials end up in landfills, increasing environmental pollution and contributing to the depletion of natural resources. Existing waste management practices primarily focus on recycling or disposal rather than encouraging reuse at the individual level.

Problem Statement

Many reusable waste materials are discarded because people lack awareness of practical and sustainable ways to transform them into useful products. There is a need for an accessible platform that can guide users by providing innovative reuse ideas, promoting environmental awareness, and encouraging responsible consumption.

Project Problem Statement

"How might we use AI to help people discover innovative ways to transform discarded materials into valuable products so that consumption and production practices become more sustainable?"

Need for the Solution

An effective solution should not only educate users about waste management but also inspire them to take action. By providing easy-to-understand recommendations and practical transformation ideas, users can reduce waste generation and contribute to environmental conservation.

NextForm addresses this need by offering an intuitive platform that identifies discarded materials, recommends suitable upcycling ideas, estimates business opportunities, and highlights the sustainability impact of each recommendation. This approach supports the principles of the circular economy while encouraging individuals to adopt environmentally responsible habits.

Target Users

The NextForm platform is designed for a wide range of users, including:

- Households interested in reducing waste and reusing everyday materials.
- Students and educational institutions promoting environmental awareness.
- Sustainability enthusiasts and environmental volunteers.
- Small businesses and entrepreneurs exploring upcycling opportunities.
- Community organizations conducting waste management and awareness programs.
- Individuals seeking creative and eco-friendly alternatives to waste disposal.

By serving diverse user groups, NextForm aims to encourage responsible consumption, promote sustainable innovation, and contribute to a cleaner and greener future.

Objectives & SDG Alignment

Primary Objective

The primary objective of **NextForm – Waste Transformation & Opportunity Platform** is to promote responsible consumption by helping users identify innovative ways to transform discarded materials into useful products. The platform aims to encourage sustainable waste management practices through AI-assisted recommendations, creating awareness about reuse, upcycling, and circular economy principles.

Specific Objectives

The specific objectives of the project are:

- To identify different categories of discarded materials such as plastic, glass, paper, cardboard, fabric, metal, wood, electronics, rubber, and organic waste.
- To recommend practical and creative upcycling ideas based on the selected waste material.
- To estimate the potential business opportunities, including approximate making cost, selling price, and profit.
- To educate users about the environmental benefits of reusing waste materials.
- To promote sustainable living and encourage responsible consumption practices.
- To provide an easy-to-use web platform that makes sustainable decision-making accessible to everyone.

Expected Outcomes

The expected outcomes of NextForm include:

- Increased awareness about waste transformation and sustainable practices.
- Reduction in the disposal of reusable materials.

- Encouragement of creative upcycling and reuse.
- Promotion of circular economy concepts within communities.
- Support for individuals and small entrepreneurs interested in developing eco-friendly products.
- Contribution towards achieving the United Nations Sustainable Development Goals.

Sustainable Development Goal (SDG) Alignment

Primary SDG

SDG 12 – Responsible Consumption and Production

NextForm primarily supports **SDG 12**, which focuses on ensuring sustainable consumption and production patterns. The platform encourages users to reduce waste by identifying reusable materials and transforming them into valuable products instead of disposing of them.

The project promotes responsible resource utilization, minimizes environmental impact, and encourages sustainable lifestyles through awareness and practical recommendations.

Secondary SDGs

SDG 11 – Sustainable Cities and Communities

By encouraging reuse and reducing waste generation, NextForm contributes to cleaner communities and more sustainable urban living.

SDG 13 – Climate Action

Reusing discarded materials helps reduce landfill waste, conserve natural resources, and lower carbon emissions associated with manufacturing new products, thereby supporting climate action initiatives.

Contribution to Sustainability

NextForm contributes to environmental sustainability by promoting the principles of the circular economy, where materials remain in use for as long as possible instead of becoming waste.

The platform empowers users to make environmentally responsible decisions by demonstrating that discarded items still possess significant value. Through AI-assisted recommendations, users gain practical knowledge about reuse opportunities, environmental benefits, and sustainable business ideas.

By integrating technology with sustainability, NextForm inspires individuals and communities to actively participate in reducing waste, conserving resources, and building a greener future.

Literature Review

Waste Management

Waste management involves the collection, transportation, processing, recycling, and disposal of waste materials in an environmentally responsible manner. With increasing urbanization and population growth, the amount of municipal solid waste generated worldwide has risen significantly. Improper disposal of waste leads to environmental pollution, health hazards, and depletion of natural resources. Sustainable waste management practices focus on reducing waste generation, promoting reuse, recycling valuable materials, and minimizing landfill disposal.

Circular Economy

The circular economy is a sustainable economic model that aims to maximize the use of resources by extending the life cycle of products through reuse, repair, refurbishment, remanufacturing, and recycling. Unlike the traditional "take-make-dispose" approach, the circular economy emphasizes keeping materials in use for as long as possible. By transforming waste into valuable resources, this model helps conserve natural resources, reduce environmental impact, and support sustainable development.

Artificial Intelligence in Sustainability

Artificial Intelligence has become an important technology for solving environmental and sustainability challenges. AI can analyze data, identify patterns, automate decision-making, and generate intelligent recommendations. In the field of waste management, AI supports material identification, waste classification, recycling optimization, and sustainable decision-making. AI-powered systems enable users to make environmentally responsible choices while improving efficiency and reducing resource wastage.

Existing Solutions

Several mobile applications and online platforms provide information about recycling and waste disposal. Some applications help users locate nearby

recycling centers, while others offer educational content on waste segregation. However, most existing solutions primarily focus on recycling rather than encouraging individuals to creatively reuse discarded materials. Many platforms also lack personalized recommendations and do not provide insights into the potential economic value of waste transformation.

Research Gap

Although various waste management solutions are available, there is limited emphasis on guiding users to transform everyday waste into useful products through simple and practical recommendations. Existing systems often overlook business opportunities, sustainability impact, and awareness of the circular economy.

NextForm addresses this gap by combining sustainability concepts with AI-assisted recommendations to help users identify reusable materials, explore innovative upcycling ideas, estimate business opportunities, and understand the environmental benefits of responsible consumption. The platform aims to make sustainable decision-making simple, accessible, and engaging for a wide range of users.

Existing System and Proposed System

Existing System

Current waste management systems primarily focus on waste collection, segregation, recycling, and disposal. While these methods help reduce environmental pollution, they often provide limited guidance on how individuals can creatively reuse discarded materials. Most existing platforms are designed to locate recycling centers or educate users about waste segregation rather than suggesting practical transformation ideas.

Additionally, many existing solutions do not provide personalized recommendations, business opportunities, or sustainability insights for different types of waste materials. As a result, reusable items such as plastic bottles, glass jars, cardboard boxes, old clothes, and metal containers are frequently discarded despite their potential value.

Limitations of the Existing System

- Limited awareness about upcycling and reuse opportunities.
- Focus mainly on recycling rather than transformation.
- Lack of personalized recommendations.
- No estimation of business opportunities from waste materials.
- Limited integration of sustainability education.
- Minimal user interaction and engagement.

Proposed System

To overcome the limitations of existing solutions, **NextForm – Waste Transformation & Opportunity Platform** has been developed as an AI-assisted web application that encourages sustainable waste management through creative reuse recommendations.

The platform allows users to enter the name of a discarded item or upload an image (future enhancement) through an intuitive interface. Based on the identified material, NextForm generates innovative upcycling ideas, estimates business opportunities, explains sustainability benefits, and aligns the

recommendations with the United Nations Sustainable Development Goals (SDGs).

The application is designed to be simple, interactive, and accessible, enabling users from different backgrounds to participate in sustainable practices without requiring technical expertise.

Advantages of the Proposed System

The proposed system offers several advantages over traditional waste management approaches:

- Encourages reuse before recycling or disposal.
- Promotes circular economy principles.
- Generates practical upcycling recommendations.
- Estimates potential business opportunities from waste materials.
- Creates awareness about environmental sustainability.
- Supports Sustainable Development Goal (SDG) 12.
- Provides a user-friendly and interactive web interface.
- Accessible from any device through a web browser.

Comparison Between Existing and Proposed System

Existing System	Proposed System (NextForm)
Focuses mainly on waste disposal and recycling	Focuses on waste transformation and reuse
Limited user interaction	Interactive web application
No personalized reuse suggestions	Provides customized upcycling recommendations
No business opportunity estimation	Estimates making cost, selling price, and profit
Limited sustainability awareness	Explains environmental impact and SDG alignment

Existing System

Recycling-oriented

Proposed System (NextForm)

Circular economy and responsible consumption oriented

5.5 Benefits of NextForm

The implementation of NextForm provides multiple environmental, social, and economic benefits.

Environmental Benefits

- Reduces landfill waste.
- Encourages responsible waste management.
- Conserves natural resources.
- Promotes sustainable lifestyles.

Social Benefits

- Increases awareness about upcycling.
- Encourages community participation in sustainability.
- Inspires students and households to practice responsible consumption.

Economic Benefits

- Highlights business opportunities from discarded materials.
- Encourages green entrepreneurship.
- Demonstrates the economic value of waste transformation.

Tools and Technologies Used

The development of **NextForm – Waste Transformation & Opportunity Platform** involved a combination of modern programming languages, web technologies, Artificial Intelligence concepts, and cloud deployment platforms. These technologies were selected to build an interactive, scalable, and user-friendly sustainability platform.

Python

Python was used as the primary programming language for developing the application. It provides a simple syntax, extensive libraries, and excellent support for Artificial Intelligence, data processing, and web application development. Python was responsible for implementing the recommendation logic, material identification, business estimation, and overall application workflow.

Streamlit

Streamlit was used to develop the web application interface. It enabled rapid development of an interactive dashboard where users can enter discarded items, upload images, and view sustainability recommendations. Streamlit simplified frontend development while maintaining seamless integration with Python.

Purpose

- User Interface Development
- Interactive Dashboard
- Data Display
- User Input Handling

IBM Granite Models

IBM Granite Models were utilized to support AI-assisted recommendation generation and sustainability-focused responses. The models helped explore intelligent reuse suggestions and practical waste transformation ideas through

prompt engineering concepts learned during the IBM SkillsBuild AI for Sustainability Internship.

Purpose

- AI-assisted recommendations
- Prompt Engineering
- Sustainability-focused responses
- Intelligent content generation

HTML

HTML was used within the Streamlit application to structure custom interface components such as information cards, headings, containers, and layout sections. It enhanced the presentation and organization of content throughout the application.

CSS

CSS was used to design the visual appearance of the application. Custom styling created a modern glassmorphism interface with responsive layouts, sustainability-themed colors, and improved user experience across different devices.

CSS Features

- Responsive Design
- Custom Buttons
- Glassmorphism Cards
- Mobile Optimization
- Sustainability Theme
- Professional Layout

Visual Studio Code

Visual Studio Code served as the primary Integrated Development Environment (IDE) for writing, testing, debugging, and managing the project. Its extensions and debugging tools improved development efficiency.

Git and GitHub

Git was used for version control, while GitHub served as the online repository for managing project files and maintaining the complete source code history. GitHub also enabled collaboration between team members and provided backup for the project.

GitHub Repository

<https://github.com/sreelekshmiaa/NextForm>

Streamlit Community Cloud

The completed application was deployed using Streamlit Community Cloud, making the project publicly accessible through a web browser without requiring local installation.

Live Website

<https://nextform-greenguardians.streamlit.app>

6.9 Technologies Summary

Technology	Purpose
Python	Core programming language
Streamlit	Web application framework
IBM Granite Models	AI-assisted recommendation generation
HTML	User interface structure
CSS	User interface styling

Technology	Purpose
Visual Studio Code	Development environment
Git	Version control
GitHub	Source code hosting
Streamlit Community Cloud	Application deployment

Advantages of the Selected Technologies

The selected technologies provide several advantages:

- Easy development and maintenance.
- Interactive and responsive user interface.
- Efficient implementation of AI concepts.
- Cloud-based deployment for global accessibility.
- Open-source development environment.
- Scalable architecture for future enhancements.
- Simple integration between frontend and backend components.

The combination of these technologies enabled the development of a reliable, user-friendly, and sustainability-focused web application capable of promoting responsible consumption and supporting the objectives of the project.

System Design and Methodology

System Overview

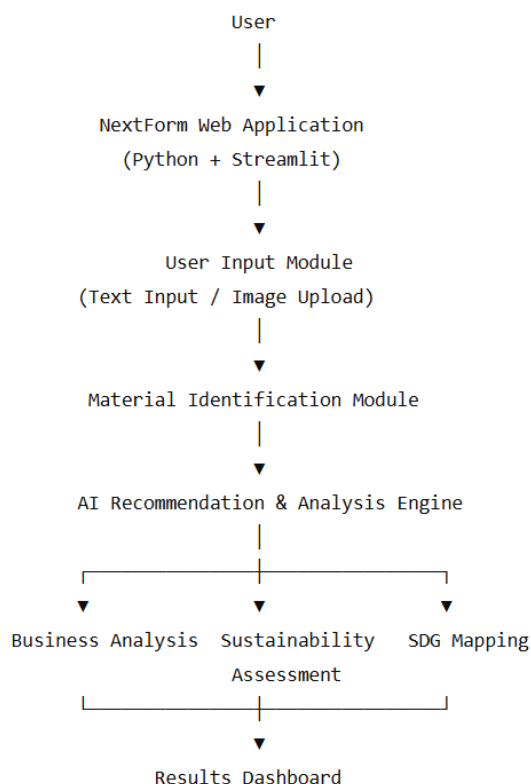
NextForm is a web-based Waste Transformation & Opportunity Platform designed to encourage sustainable waste management through AI-assisted recommendations. The system allows users to enter the name of a discarded item and receive practical suggestions for transforming it into useful products. Along with reuse recommendations, the platform provides business opportunity estimates, sustainability impacts, and Sustainable Development Goal (SDG) alignment.

The application has been developed using Python and Streamlit with a user-friendly interface that makes sustainable decision-making simple and accessible.

System Architecture

The architecture of NextForm consists of several interconnected modules that work together to process user input and generate meaningful sustainability recommendations.

System Architecture



System Modules

1. User Input Module

This module allows users to enter the name of a discarded item or upload an image. The input serves as the starting point for the recommendation process.

2. Material Identification Module

The entered item is analyzed to determine its material category, such as plastic, glass, paper, cardboard, fabric, metal, wood, electronics, rubber, or organic waste.

3. Recommendation Engine

Based on the identified material, the recommendation engine generates innovative upcycling ideas and practical reuse suggestions. It also identifies the most suitable transformation option for the selected item.

4. Business Opportunity Module

This module estimates:

- Approximate making cost
- Expected selling price
- Estimated profit

These estimates encourage users to explore entrepreneurship opportunities through sustainable waste transformation.

5. Sustainability Assessment Module

The system explains how each recommendation contributes to environmental sustainability by reducing landfill waste, conserving natural resources, and encouraging responsible consumption.

6. SDG Alignment Module

NextForm maps every recommendation to the relevant Sustainable Development Goal, primarily **SDG 12 – Responsible Consumption and Production**, helping users understand the global significance of their actions.

Methodology

The development of NextForm followed a structured methodology inspired by the Design Thinking process.

Step 1 – Problem Identification

A real-world sustainability challenge was identified by observing the increasing amount of reusable waste generated in households and communities.

Step 2 – Requirement Analysis

The project requirements were analyzed to determine user needs, sustainability goals, and suitable AI concepts for the solution.

Step 3 – System Design

A modular architecture was designed to separate user input, material identification, recommendation generation, business analysis, and sustainability assessment.

Step 4 – Development

The application was implemented using Python and Streamlit. HTML and CSS were incorporated to create an attractive and responsive user interface.

Step 5 – Testing

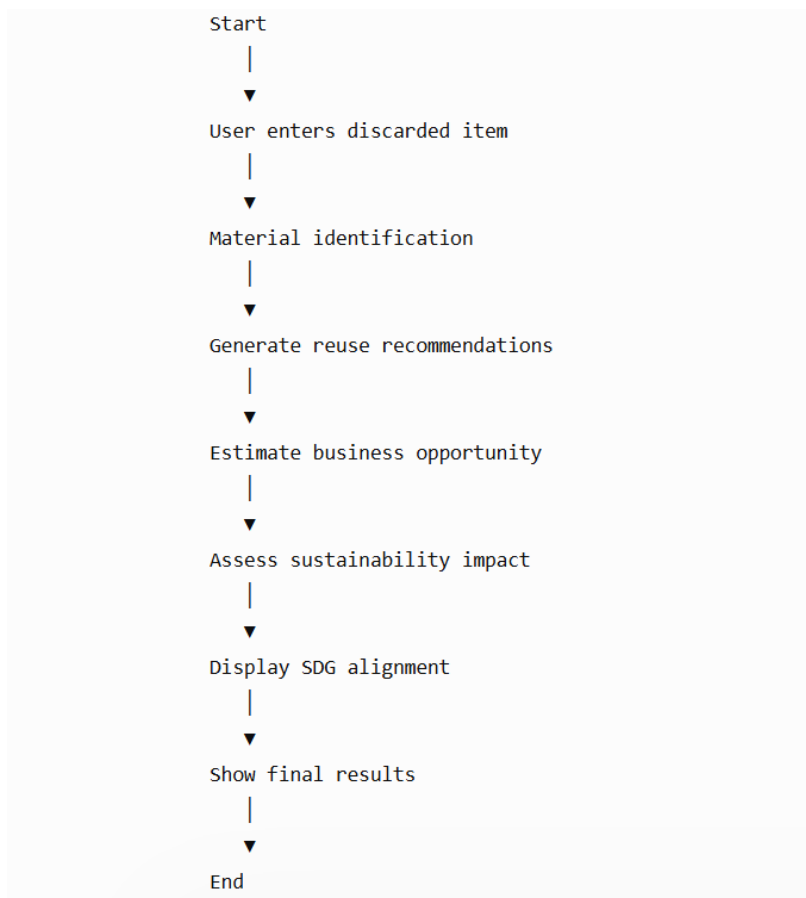
Different waste items were tested to verify the accuracy of recommendations, business estimates, and sustainability outputs.

Step 6 – Deployment

The completed application was deployed on Streamlit Community Cloud, making it publicly accessible through a web browser.

Workflow

The overall workflow of NextForm is illustrated below:



Advantages of the Proposed Design

The modular design of NextForm offers several advantages:

- Easy to maintain and update.
- Scalable architecture for future AI enhancements.
- User-friendly workflow.
- Efficient recommendation process.
- Supports cloud deployment.
- Promotes responsible and sustainable decision-making.

The architecture ensures that each component performs a specific function while working together to provide users with meaningful and practical sustainability recommendations.

Results and Discussion

Project Results

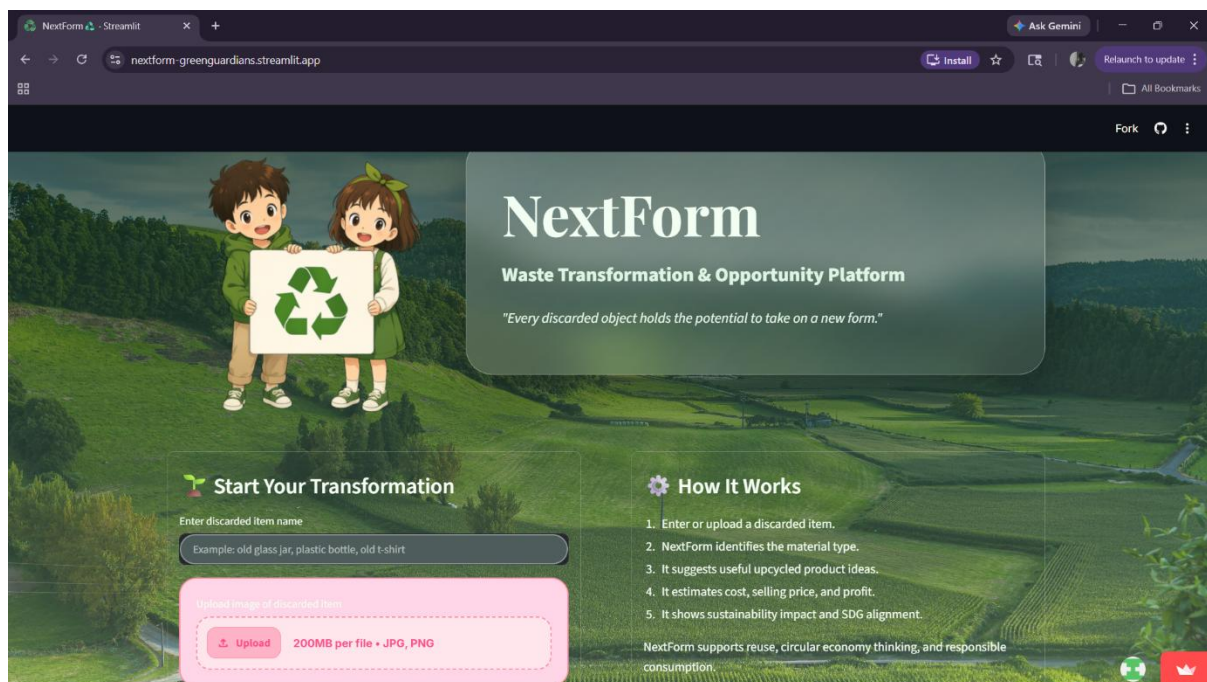
The development of **NextForm – Waste Transformation & Opportunity Platform** resulted in a fully functional web application that helps users discover innovative ways to transform discarded materials into useful products. The platform provides practical recommendations, business opportunity estimates, sustainability insights, and Sustainable Development Goal (SDG) alignment through an intuitive and user-friendly interface.

The application was successfully deployed using **Streamlit Community Cloud**, allowing users to access the platform from any device with an internet connection.

Live Website

<https://nextform-greenguardians.streamlit.app>

Home Page



The home page introduces the NextForm platform and provides users with a simple interface to enter the name of a discarded item or upload an image. The interface has been designed with a sustainability theme and responsive layout to enhance user experience.

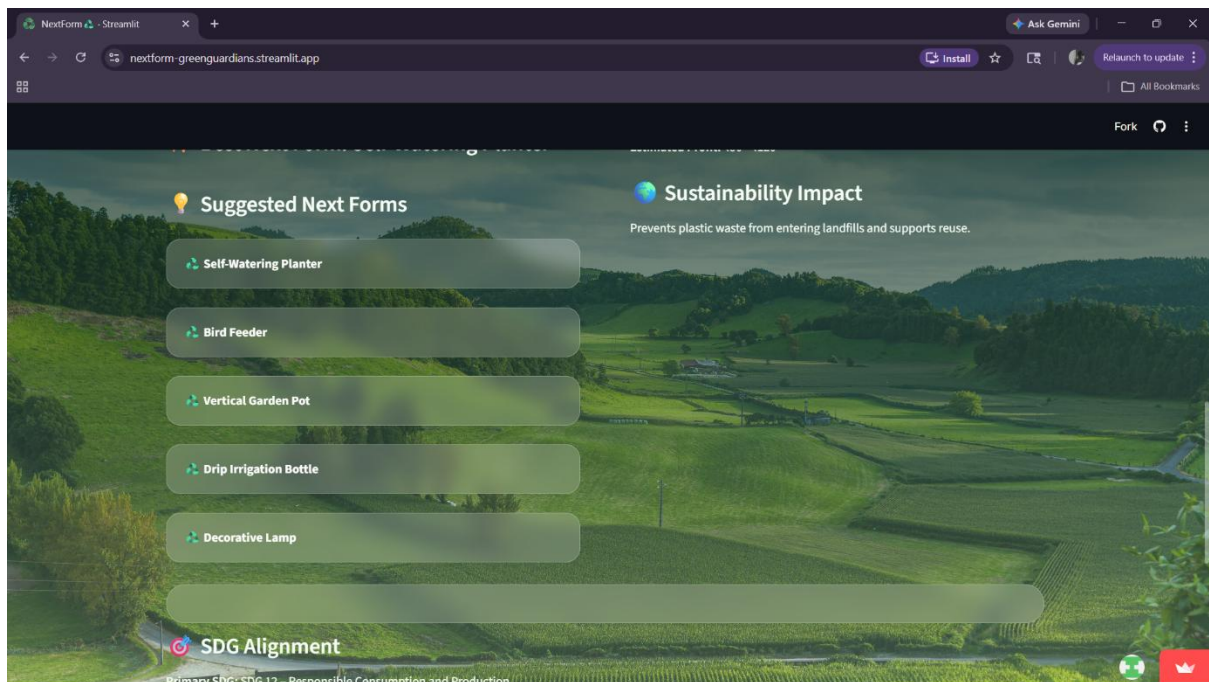
Material Identification

The screenshot shows the NextForm web application interface. The background is a scenic landscape with green hills and a winding road. At the top, there's a navigation bar with the NextForm logo, a Streamlit link, and a Gemini icon. Below the navigation bar, there's a header section with a large image of two people standing on a hill. The main content area is divided into two columns. The left column is titled 'Start Your Transformation' and contains a form with a text input field labeled 'Enter discarded item name' containing the text 'plastic bottle', a 'Press Enter to apply' button, an 'Upload image of discarded item' section with an 'Upload' button and a note '200MB per file • JPG, PNG', and a green button labeled 'Analyze Item'. The right column is titled 'How It Works' and contains a list of five steps: 1. Enter or upload a discarded item. 2. NextForm identifies the material type. 3. It suggests useful upcycled product ideas. 4. It estimates cost, selling price, and profit. 5. It shows sustainability impact and SDG alignment. Below the list, it says 'NextForm supports reuse, circular economy thinking, and responsible consumption.'

The screenshot shows the NextForm web application interface displaying the results of an analysis. The background is the same scenic landscape. At the top, there's a green banner with a checkmark and the text 'Analysis Complete!'. Below this, the 'NextForm Analysis' section displays four metrics: Reusability (90/100), Profitability (78/100), Sustainability (92/100), and NextForm Score (87/100). Each metric has a corresponding progress bar. Below the metrics, there are four sections: 'Item Understanding' with a blue icon, 'Detected Material: Plastic' with a blue icon, 'Best Next Form: Self-Watering Planter' with an orange icon, and 'Suggested Next Forms' with a yellow icon. The 'Suggested Next Forms' section shows a 'Self-Watering Planter' option. To the right, there's a 'Business Opportunity' section with a green icon, showing 'Estimated Making Cost: ₹20 - ₹60', 'Estimated Selling Price: ₹60 - ₹180', and 'Estimated Profit: ₹50 - ₹120'. Below this is a 'Sustainability Impact' section with a blue icon, stating 'Prevents plastic waste from entering landfills and supports reuse.'

When a user enters a discarded item, the system identifies its material category, such as plastic, glass, paper, cardboard, fabric, metal, wood, electronics, rubber, or organic waste. This classification serves as the foundation for generating suitable reuse recommendations.

Upcycling Recommendations



Based on the identified material, NextForm generates innovative transformation ideas that encourage reuse instead of disposal.

Example

Input: Plastic Bottle

Detected Material: Plastic

Suggested Next Forms:

- Self-Watering Planter
- Bird Feeder
- Vertical Garden Pot
- Decorative Lamp
- Drip Irrigation Bottle

These recommendations encourage users to extend the lifecycle of waste materials while reducing environmental impact.

Business Opportunity Analysis

The platform estimates the potential economic value of transforming waste materials by providing:

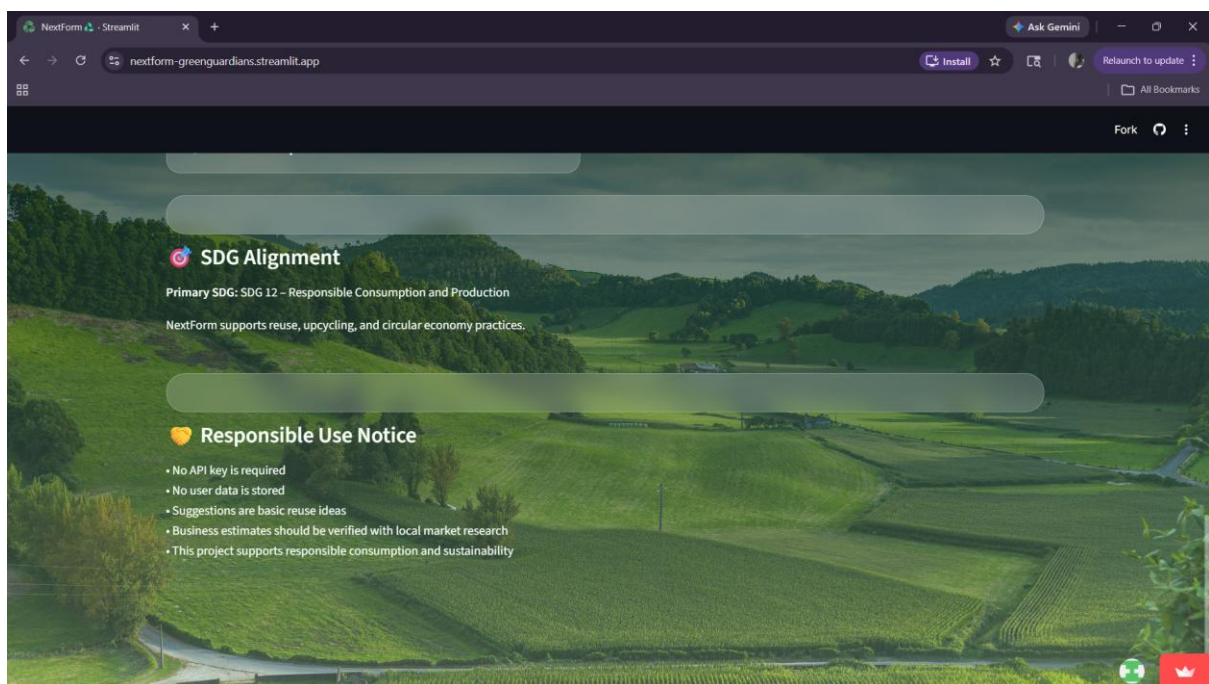
- Estimated Making Cost
- Estimated Selling Price
- Estimated Profit

These estimates help users understand that sustainable practices can also create entrepreneurial opportunities.

Sustainability Impact

NextForm explains how each recommendation contributes to environmental sustainability. The platform highlights the benefits of reducing landfill waste, conserving natural resources, and promoting responsible consumption.

SDG Alignment



The platform maps recommendations to **Sustainable Development Goal 12 Responsible Consumption and Production**, helping users understand the global significance of their sustainable actions.

Testing Results

The application was tested using different types of waste materials to verify the recommendation engine and overall functionality.

Input Item	Detected Material	Sample Recommendation
Plastic Bottle	Plastic	Self-Watering Planter
Glass Jar	Glass	Candle Holder
Cardboard Box	Cardboard	Desk Organizer
Old T-Shirt	Fabric	Tote Bag
Tin Can	Metal	Pen Stand
Coconut Shell	Organic	Bird Feeder

The system successfully generated appropriate recommendations and sustainability insights for all tested items.

Discussion

The results demonstrate that NextForm effectively promotes responsible consumption by helping users recognize the value of discarded materials. The combination of sustainability awareness, practical recommendations, and business opportunity estimation encourages users to adopt environmentally responsible practices.

The project successfully integrates AI concepts with sustainable development principles, making waste transformation simple, accessible, and engaging. Furthermore, the responsive web interface and cloud deployment ensure that the platform can be accessed by users across different devices and locations.

Responsible AI, Challenges Faced, Learning Outcomes, Future Scope and Conclusion

Responsible AI Considerations

Artificial Intelligence has the potential to solve real-world sustainability challenges, but it must be developed and used responsibly. During the development of NextForm, the principles of Responsible AI introduced through the IBM SkillsBuild AI for Sustainability Internship were considered throughout the design process.

The following principles were incorporated into the project:

Fairness

The platform is designed to provide recommendations based on the characteristics of discarded materials rather than personal information. Recommendations are generated consistently without bias toward any individual or group.

Transparency

NextForm clearly presents the detected material, recommended transformations, estimated business opportunities, and sustainability impact. Users can easily understand how recommendations are generated.

Privacy

The application does not collect, store, or share users' personal information. User inputs are processed only for generating recommendations, ensuring privacy and responsible handling of data.

Human Oversight

The recommendations provided by NextForm are intended to support users in making informed decisions. Final decisions regarding reuse, implementation, or business opportunities remain with the user.

Environmental Responsibility

The primary purpose of the platform is to encourage responsible consumption, reduce waste generation, promote reuse, and support sustainable development through practical recommendations.

Challenges Faced

During the development of NextForm, several technical and design challenges were encountered.

Technical Challenges

- Understanding Streamlit framework as beginners.
- Designing an attractive and responsive user interface.
- Integrating AI concepts into the recommendation workflow.
- Managing project files using Git and GitHub.
- Deploying the application successfully using Streamlit Community Cloud.

Project Challenges

- Selecting an appropriate sustainability problem.
- Mapping the solution to Sustainable Development Goal 12.
- Designing meaningful recommendations for different waste materials.
- Balancing simplicity with useful functionality.

Each challenge provided valuable learning opportunities and improved our technical and problem-solving skills.

Learning Outcomes

The internship and project development significantly enhanced both technical and professional competencies.

Technical Skills

- Python Programming
- Streamlit Web Application Development
- Git and GitHub Version Control
- Website Deployment using Streamlit Community Cloud
- HTML and CSS Customization
- AI Concepts and Prompt Engineering

- Sustainable Technology Development

Professional Skills

- Design Thinking
- Problem Solving
- Team Collaboration
- Project Documentation
- Presentation Skills
- Responsible AI Awareness
- Sustainable Innovation

Future Scope

Although NextForm successfully demonstrates the concept of AI-assisted waste transformation, several enhancements can further improve the platform.

Future developments may include:

- Real-time image recognition for automatic material detection.
- Advanced AI-powered recommendation models.
- Mobile application development for Android and iOS.
- Community platform for sharing upcycling ideas.
- Marketplace for buying and selling upcycled products.
- Personalized sustainability recommendations based on user preferences.
- Integration with local recycling centers and municipal waste management systems.
- Multilingual support to increase accessibility.

These enhancements would improve user experience while expanding the platform's contribution to environmental sustainability.

Conclusion

NextForm – Waste Transformation & Opportunity Platform demonstrates how technology can be used to encourage sustainable living and responsible consumption. By combining Artificial Intelligence concepts with practical waste transformation ideas, the platform helps users recognize the hidden value of discarded materials and promotes environmentally responsible decision-making.

The project successfully aligns with **United Nations Sustainable Development Goal 12 – Responsible Consumption and Production** by encouraging reuse, reducing waste generation, and supporting circular economy principles.

The development of NextForm provided valuable experience in software development, web application design, AI-assisted problem solving, cloud deployment, and sustainable innovation. Beyond building a functional application, the project strengthened our understanding of how technology can create meaningful environmental and social impact.

We believe that NextForm has the potential to inspire individuals, educational institutions, and communities to adopt sustainable practices and view waste not as a problem, but as an opportunity for innovation and positive change.

References

1. IBM SkillsBuild – AI for Sustainability Learning Resources.
2. AICTE – Internship Guidelines and Resources.
3. 1M1B Foundation – AI for Sustainability Virtual Internship.
4. United Nations. *Sustainable Development Goals (SDGs)*.
5. Python Software Foundation. *Python Documentation*.
6. Streamlit Documentation.
7. GitHub Documentation.
8. Circular Economy Principles – Ellen MacArthur Foundation.

Appendix

Appendix A – Live Website

<https://nextform-greenguardians.streamlit.app>

Appendix B – GitHub Repository

<https://github.com/sreelekshmiaa/NextForm>

Appendix D – Project Screenshots

Attach screenshots of:

- Home Page
- User Input Interface
- Material Detection
- Upcycling Recommendations
- Business Opportunity Analysis
- Sustainability Impact
- SDG Alignment
- Mobile Responsive View

Appendix C – Source Code Structure

NextForm/

- |— app.py
- |— requirements.txt
- |— README.md
- |— recycle.png
- |— greenlove.png
- |— assets/

Appendix D – Deployment Details

- **Development Environment:** Visual Studio Code
- **Programming Language:** Python
- **Framework:** Streamlit
- **Version Control:** Git & GitHub
- **Deployment Platform:** Streamlit Community Cloud

Appendix E – Team Details

Project Title: NextForm – Waste Transformation & Opportunity Platform

Team Name: Green Guardians

Team Members:

- Sreelekshmi A A
- Sooryajith S

Internship: 1M1B AI for Sustainability Virtual Internship

In Collaboration With: IBM SkillsBuild & AICTE

